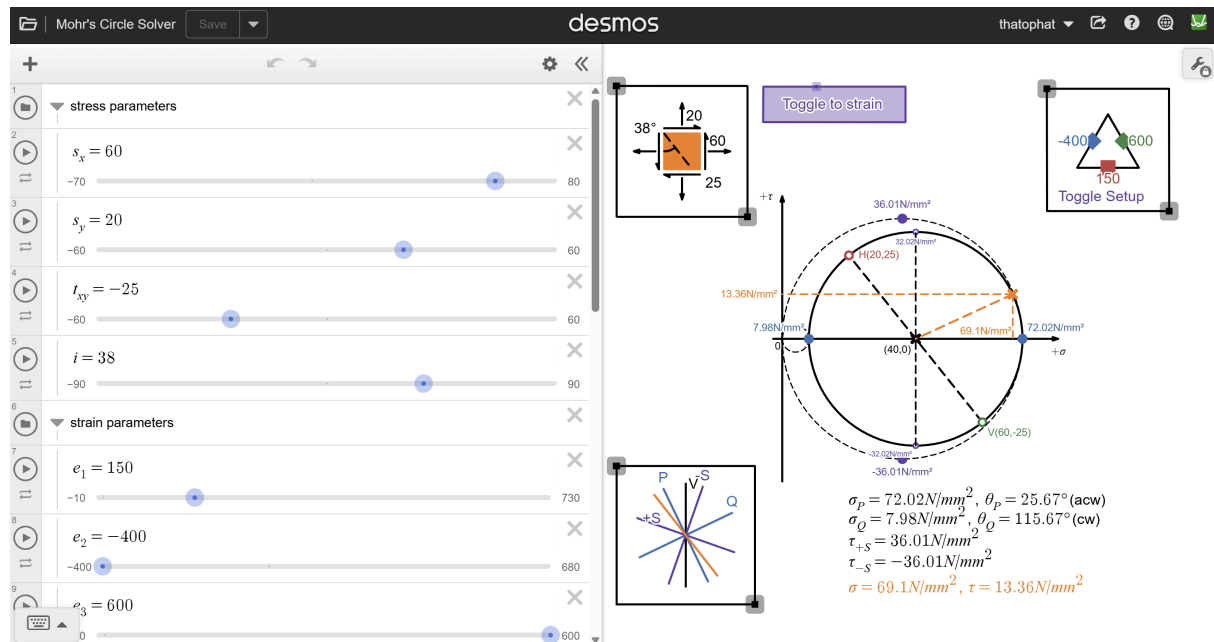


MECH3 Mohr's Circle Solver

the solver:

<https://www.desmos.com/calculator/dtc55god8k>



What does this do?

- This calculator has two modes, one mode solves for the principal stresses, maximum shear, and stresses of a joint given the planar stress state, while another mode solves for the principal strains and the strain circle given the strain state of a strain gauge rosette.

What are the limits of this?

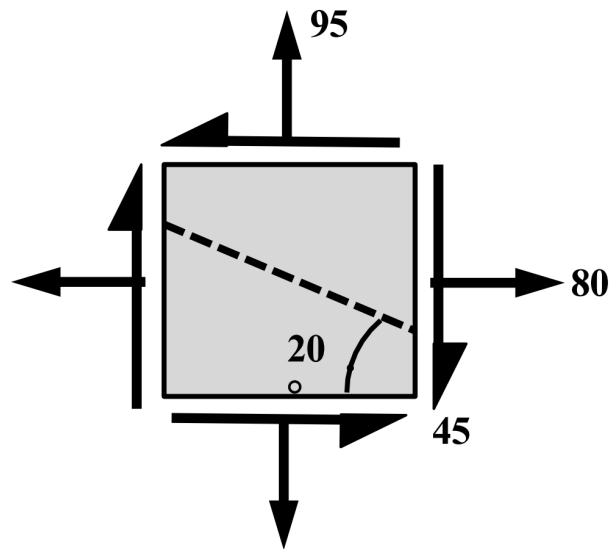
- Only planar stresses are computed, meaning that it is assumed that the stress in the z direction is 0. It also mainly works when 0 is not a main principal stresses, as it fails to render the Mohr's circle properly when the two principal stresses are close to each other. The strain circle also only comes in 3 setups, and cannot directly give the strain in one direction given the strain state.

How do i use it?

There are two modes of this calculator:

Mohr's Stress Circle

Say we had to find the principal stresses, maximum shear, and stresses at the joint of the given stress state:



1. Input the applied Stresses

The element is subjected to the direct stresses $\sigma_x = 80\text{N/mm}^2$, $\sigma_y = 95\text{N/mm}^2$ and a shear stress of $\tau_{xy} = 45\text{N/mm}^2$.

Thus, type 80, 95 and 45 in the respective sliders here:

1		▼ stress parameters	×
2		$s_x = 80$	×
		-70 80	
3		$s_y = 95$	×
		-60 95	
4		$t_{xy} = 45$	×
		-60 60	

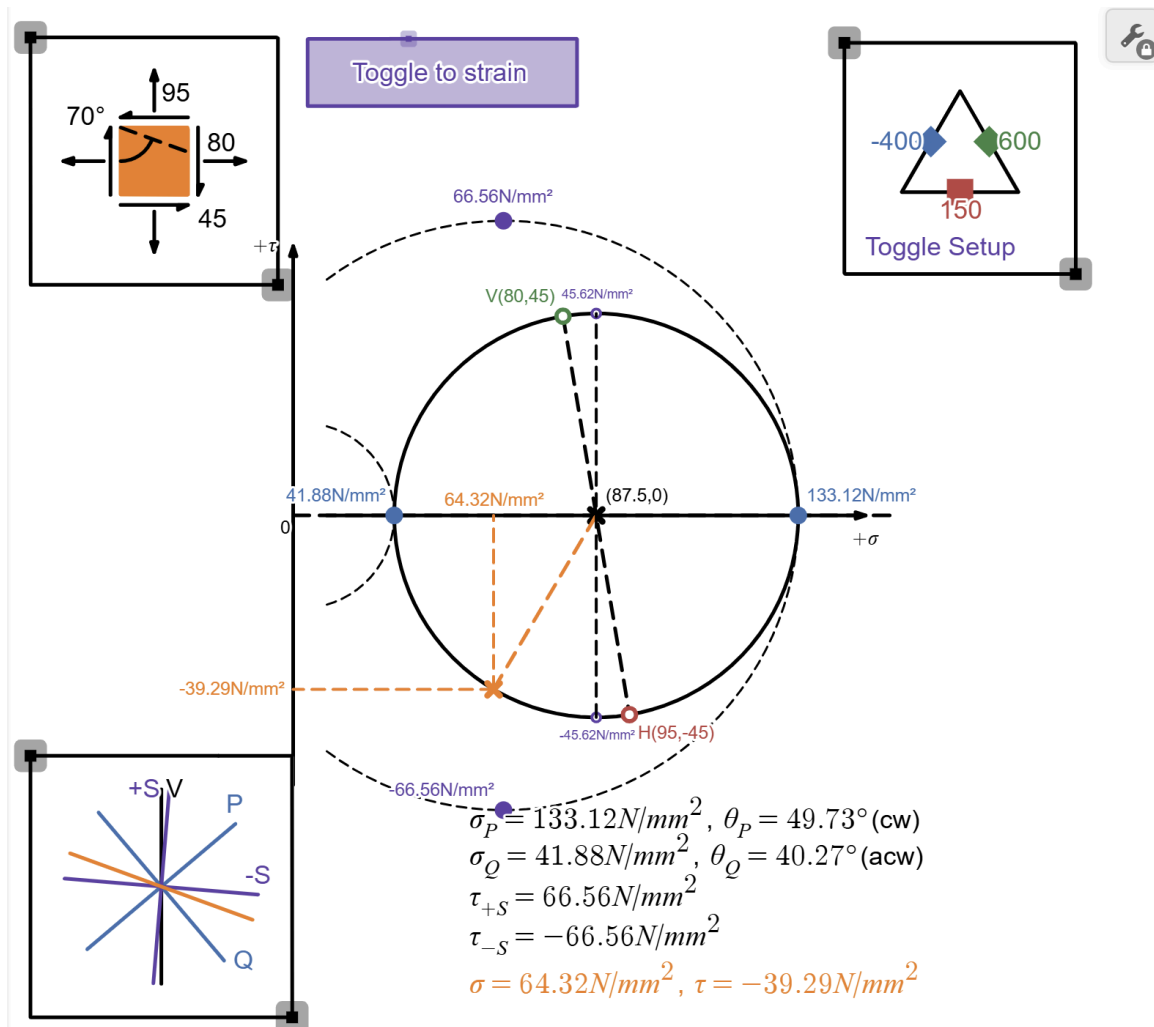
2. Input the Joint Angle

The joint in the element is placed 20 degrees clockwise from the horizontal. However, the solver only takes in values from the vertical plane. Thus, the joint is actually $90^\circ - 20^\circ = 70^\circ$ from the horizontal, counterclockwise. Since the solver has counterclockwise angles as positive, enter 70 here:



3. Read off the Values

Now, you should see the element in the top left corner match with your stress state, and the Mohr's circle plotted with all the readings.



On the top left corner is the stress element, that should match up with the one given. In the middle is the stress circle, with the V and H points plotted, alongside the other coordinates, like the center of the circle, the principal stresses, the maximum stresses, and the joint. The solver also automatically draws the other planes' circles, if the stress in the z direction becomes a principal stress.

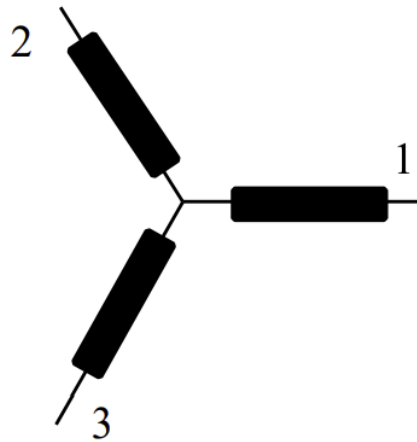
On the bottom is the numerical values for the principal stresses and maximum shears, as well as the angle of their planes from the vertical plane. The joint stresses are also displayed there too. The bottom left corner shows a diagram of all the major planes' inclinations from the vertical plane.

Thus, the values are

$$\begin{aligned}\sigma_P &= 133.12MPa, \sigma_Q = 41.88MPa \\ \tau_{max} &= \pm 66.56MPa \\ \sigma_J &= 64.32MPa, \tau_J = -39.2MPa\end{aligned}$$

Mohr's Strain Circle

Say we had to find the principal strains of the strain state of the below strain gauge rosette:



(b) Star

$$\epsilon_1 = 730 \times 10^{-6}$$

$$\epsilon_2 = 680 \times 10^{-6}$$

$$\epsilon_3 = 190 \times 10^{-6}$$

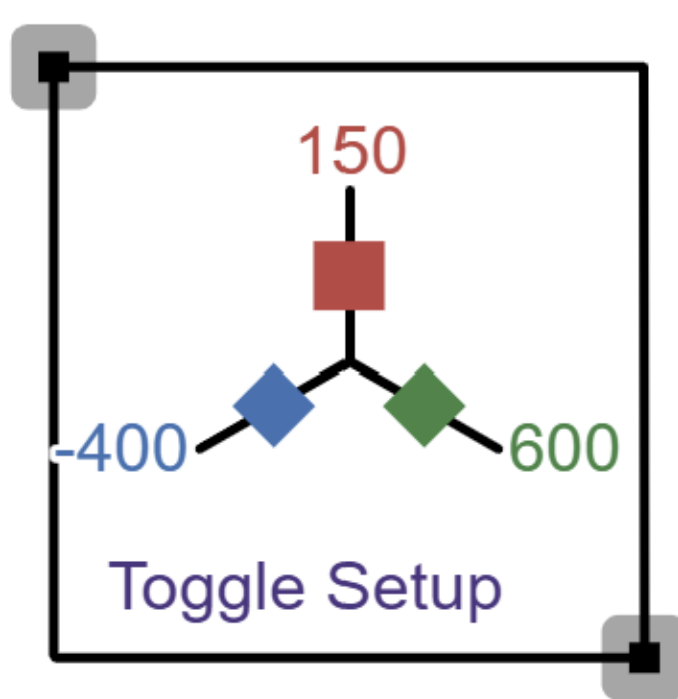
1. Toggle to the strain mode and select the correct rosette

First, toggle to the strain mode by clicking the strain button at the top:



The circle should now change.

Then, toggle the rosette by clicking the purple text to reach the correct rosette setup.



The solver only supports rectangular, star, and delta rosettes.

2. Input the strain gauge values

The strain gauge window shows three gauges with **gauge 1 being red**, **gauge 2 being blue**, and **gauge 3 being green**.

Since the order of the gauges is counterclockwise, which is the same as the given rosette, we can copy the values directly. Thus, input

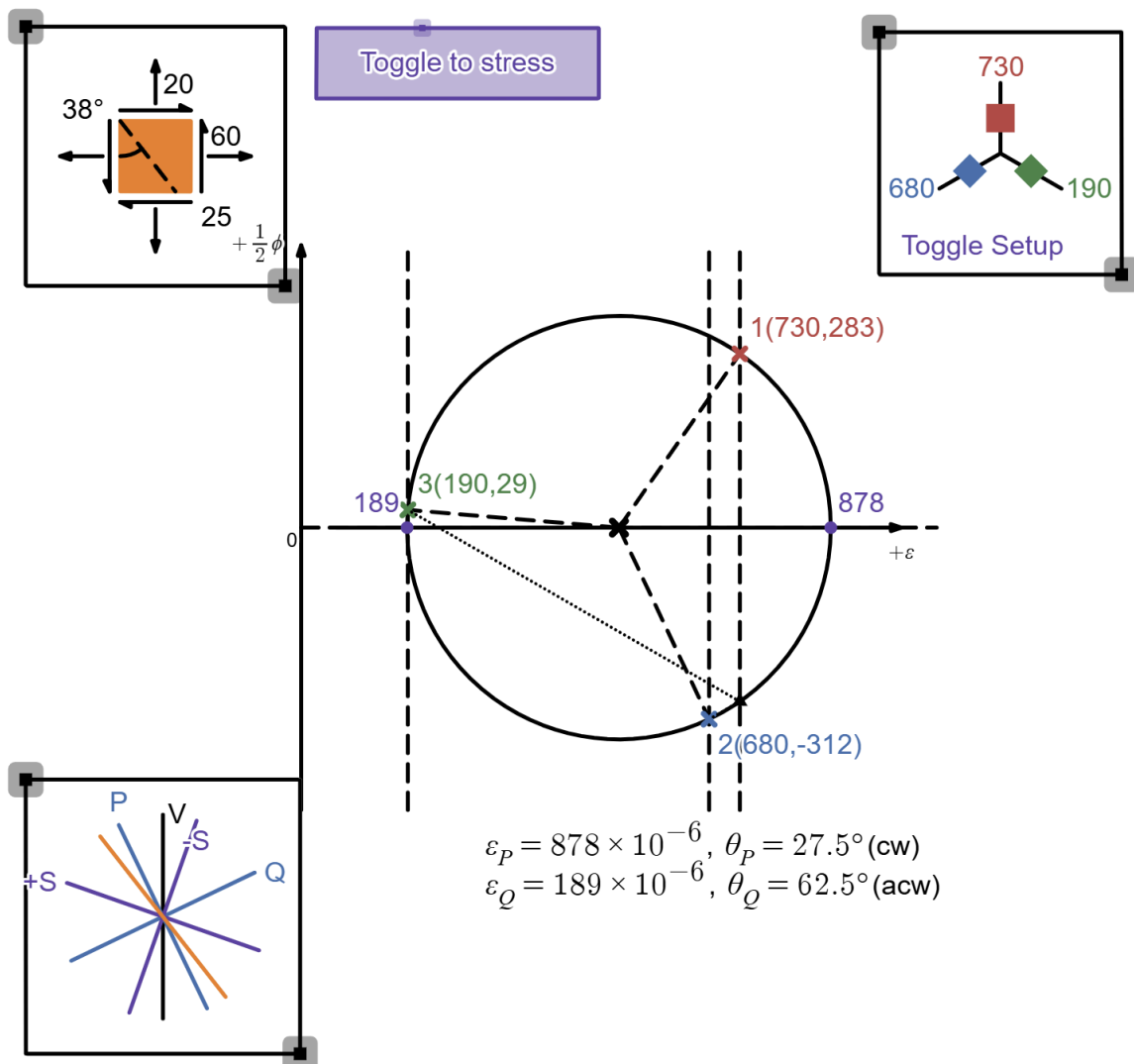
$$e_1 = 730, e_2 = 680, e_3 = 190$$

strain parameters		
6		
7		$e_1 = 730$
		-10 730
8		$e_2 = 680$
		-100 680
9		$e_3 = 190$
		-10 200

If the order of the strain gauges are different than the solver, change the values to match the given order.

3. Read the values

The strain gauge rosette should now be updated with the correct values, and the Mohr's Strain Circle should be plotted.



The circle now shows the three points corresponding to the gauges, the principal stresses, and the construction lines for the traditional McClintock's Method.

On the bottom, the values for the principal strains are shown, along with the angle of the direction from gauge 1.